JP30Words: An interactive Learning Website.

Overview

The set goal was to create a website that is a self-guided learning tool set around 30 Japanese words. The categories of Nouns, Verbs, and Adjectives are present with dictionary links. Along with a small guide on how the test works. Navigation made easy on desktop, tablet, and mobile. Entries are concise and to the point. And external links are provided for further learning opportunities.

So What is New?

The first immediate deviation from the plan that I wish I thought of when I wrote it, was a section for expressions that are -not- part of the test. The expressions would be more challenging, so they still were not included in the test game, but are there to enlighten or even embolden the user who may be familiar with them from Japanese media consumption.

The second deviation was some unforeseen multi-platform changes. The navigation menu I conceived of with an arrow button seems good for mobile devices, but on desktop it was a little bit unnecessary to add an extra click on the far side of the screen. So a different desktop and potentially more tablet friendly menu is placed besides the main body of text when the screen is wide enough. And the arrow button is hidden.

Target Audience Appeal

When it comes to the audience of potential Japanese learners and Anime enthusiasts. They will find the ease of access and use of the test on any device, and the links provided in the resource tab will be bread crumbs to their destination of Japanese literacy.

Core Features

- Header with logo image at the top of the page. (HTML, CSS)
- Body and Navigation beside each other on different rows. (HTML, CSS)
- Navigation button and menu for small screens. With smooth animation. Being placed beside the main body to reduce clicks when screen width is large enough. (HTML, CSS, JavaScript.)
- Navigation menus jump to each section (HTML.)
- Page with the test/guessing game. Drawn onto a canvas with buttons to interact. (HTML, CSS, JavaScript.)
- The test page has a list of four different tests: Nouns, Adjective, Verbs, and Everything. (HTML, CSS, JavaScript.)
- Each deck gets shuffled when you click the canvas to start the test. So the order of appearance of the words is random. This extends to all of the projects JP30 words. This uses 4 sets of 4 arrays that make up the cards contents. The word with Kanji, just Hiragana, converted to Romanji, and the word in English.
- The English word and the Hiragana are included on the card by default, but a checkbox exists for each respectively, if unchecked, it will hide each element. Meaning English can be turned off (Internally I refer to this as a ‘hint’) while the Hiragana is intended and is only there as an option to test against the Kanji skills.
- Resources section with additional learning opportunities. (HTML)s

Design/UX Considerations

For the visual experience of the website, the basic premise was simple tables, a header, and sections. With a footer for copyright. The navigation menu and multi-platform considerations is what really changed how I developed the websites appearance from a bounds standpoint. Originally I didn’t imagine a desktop version needing the separate navigation menu, but it was a much more intuitive idea when I got to that point in development.

When it comes to color choice I ended up picking crimson, white, lavender, pink, and an off pink. The accents ended up being much more subtle than in my mind. The unplanned off pink works better when I used it for the linear gradient on the tables data. Though hovering over a table elements data point uses the pink as part of the highlight effect on the modified linear gradient.

When it came to the test page, it was a bit more complicated than I expected with the footer. It actually would bounce around the page. I ended up making it fixed to the bottom left and gave it a very small less intrusive size to not impede usability.

As for the test itself, the way the buttons needed to be laid out just took testing. Adding in pressing enter to input the answer made it much more enjoyable when using a keyboard.

Challenges

Nearly every part of the website had me asking questions that my initial plan didn't fully account for. Somehow in my mind I was gonna make navigation work without JavaScript, which changed quickly. On that topic. The page layout itself, the margins, had to change with the screen size.

The header stacking over the descriptor text if the screen is just that small to avoid a break in the style sheet was one challenge for mobile devices. Large screens needed their own menu and it was initially two navigation menus that worked, but was messy, until I thought out a better process.

When it came to the test. I had to conceive of the A to B to C roughly following the original plan using the JavaScript technologies I had just learned. I've worked most with arrays in other languages, so I opted out of using JSON, which could've been the other way to make these cards. I used a canvas to make the test more visually appealing, though I still kept it very simple within that context. I referred to the study material and internet searches when I was lost or needed context for things I know I could do, but forgot the syntax for.

Speaking of, switching between HTML, CSS, and JavaScript certainly had it share of syntax issues. It takes extra awareness and testing to make sure it is all working as intended, especially with any changes between multiple files. Even working incrementally there was sometimes where things would act unexpected and I had to retrace my steps to even before the change to find where I made a logical error.

Lastly was time management. I felt overwhelmed with work overall so most of the project was not completed on the timeline. I got just part of the HTML done before the final couple weeks and kicked it into high gear. It is not best practice, but I'm a student who is learning and struggling sometimes to stay focused despite my best efforts over multiple ongoing classes while balancing it with work. I'm not trying to rely on 'pressure', but I certainly relied on a reduced workload to spend my time pushing development forward. I endeavor to improve as a student and believe this is part of the growing pains.

Lessons Learned

- **Planning is important;** Despite having much less information about JavaScript at the time, I still refereed to my proposal to keep a checklist of tasks to push development forward.
- **Plans change;** Where I expected feature creep ended up being in different places. The test I added the checkboxes for settings visible elements on the card, while the introduction and references were more sparse.
- **Trial and error is inevitable;** The Design/UX was definitely not set in stone by the proposal. I was able to mitigate choice paralysis, but the process of simply looking and feeling if the design is right is an art.
- **Building a project is motivating;** After all of this, I'm excited to do my own projects in my own time.